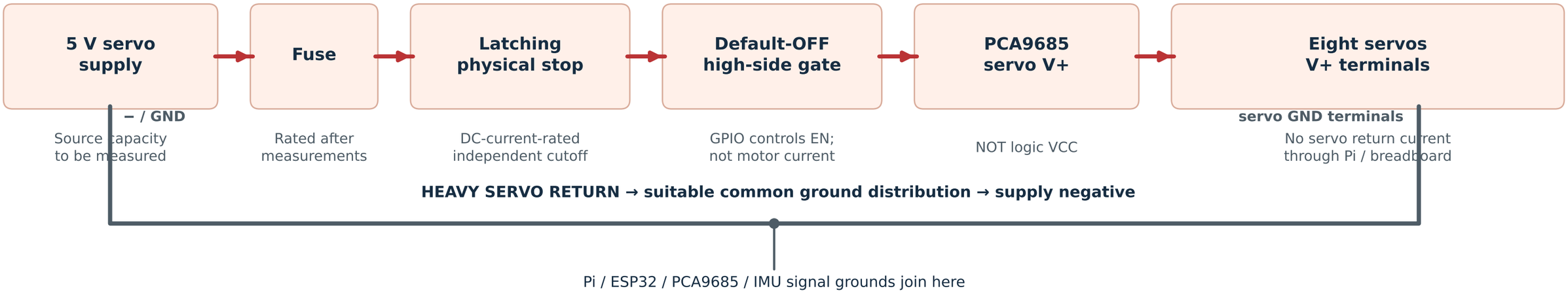
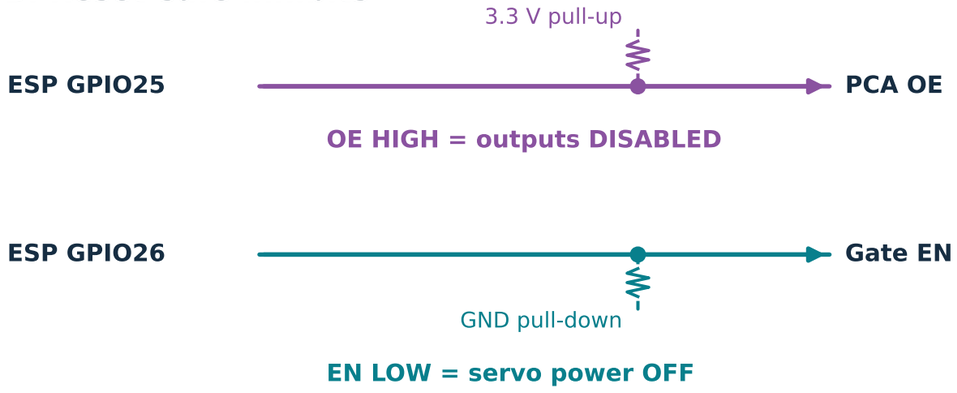
04 / SERVO POWER, INHIBITS & CHANNEL MAP

Planning diagram only. The current firmware has no PWM backend; never energize servos to try the software demo.

A. Dedicated servo positive supply — separate from the Pi/ESP32 positive rail



B. Reset-safe inhibits



External resistors required; values/circuit depend on selected boards.

C. PCA9685 signal channel → joint

CH 0	left_hip_pitch	LEG
CH 1	left_knee	LEG
CH 2	left_ankle_pitch	LEG
CH 3	right_hip_pitch	LEG
CH 4	right_knee	LEG
CH 5	right_ankle_pitch	LEG
CH 6	head_pan	HEAD
CH 7	head_tilt	HEAD

Left/right means the robot's own left/right, not the viewer's.

D. Identify the three actual servo connections before plugging in

SIGNAL → assigned CH **V+ → switched servo 5 V** **GND → heavy return**

Header order and servo lead colors vary: use board labels and the servo datasheet. Do not infer pin order from this sheet.

Before any future output: measured calibration, rated power/stop/gate, supported mechanism and commissioning checks are required.